Software Requirements Specification

for

Performance Monitoring Systems

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for the ***Performance Monitoring System****, version 1.0.*
The system is designed to track, monitor, and analyze the academic performance of students under the supervision of Faculty Advisors or Supervisors.

1.2 Document Conventions

This SRS follows ***IEEE Software Requirements Specification standards****.*
All functional requirements are stated using clear, verifiable statements in the form ***"The system shall…"****.*
Each requirement is uniquely identified using requirement IDs (e.g., FR-STU-01).
Priority levels are expressed as ***High, Medium, or Low*** *where applicable.*
Important points and terms have been expressed in ***Bold*** *to provide emphasis.*
Abbreviations are used in some places.

1.3 Intended Audience and Reading Suggestions

The intended audience is the team of developers who will be designing and implementing the Property Management System. Also, the document is to be utilized by the testing team who will be testing and evaluating the performance and design of the application. The document consists of all the necessary information that will be required by the team of software engineers who will be working on the project.

Readers are advised to begin with ***Section 1 and Section 2*** *for an overview and then proceed to* ***Section 4 (System Features)*** *for detailed functional requirements.*

1.4 Product Scope

The Performance Monitoring System is a ***centralized web-based academic platform*** *that supports three user roles:*

- *Student*
- *Faculty Advisor / Supervisor*
- *Admin*

The system integrates academic data management, verification workflows, feedback mechanisms, reporting, alerts, and guidance tools into a single platform. It aims to simplify academic monitoring, improve transparency, and reduce manual overhead for both students and faculty.To make it accessible on all electronic devices the application will be a web-based application and will be mobile responsive as much as possible.

1.5 References

Websites :

- *React https://react.dev/learn*
- *SpringBoot https://docs.spring.io/spring-boot/documentation.html*
- *PostgreSQL https://www.postgresql.org/docs/*

2. Overall Description

2.1 Product Perspective

*The Performance Monitoring System is a **brand-new, self-contained web application**.*
It is not a replacement or extension of any existing system and operates independently within a standard web environment. The system intends to provide an alternative and easier way to manage academic data verification workflows, feedback mechanisms, reporting, alerts, and guidance tools into a single platform.

2.2 Product Functions

System provides the following functions:

- *User registration and authentication*
- *Academic data entry and monitoring*
- *Verification and approval workflows*
- *Automated alerts and email notifications*
- *Academic report generation*
- *Chatbot-based academic guidance*
- *Attendance tracking*
- *Text extraction from academic documents*

2.3 User Classes and Characteristics

User Class	Technical Expertise	Privilege Level	Usage Frequency
Admin	High	Maximum	Moderate
Student	Low	Low	High
Faculty Advisor / Supervisor	Low	Moderate	Moderate

2.4 Operating Environment

*The system operates as a **web-based application** accessible through standard modern web browsers. Since the application is a web application it can work on any device having a browser.*

- *Device: Mobile Phone, Computer, Laptops, Tablets.*
- *Operating System: Windows, Linux distributions, Mac OS, Android*
- *RAM: 128 MB or more*

2.5 Design and Implementation Constraints

- *Backend framework: **Spring Framework***
- *Frontend framework: **React***
- *Database: **PostgreSQL***
- *Architecture: Web-based client–server mode*

2.6 User Documentation

Appropriate instructions will be provided at every step in the application to ensure the users do not face any difficulties while using the application.Appropriate error messages will be shown in case the user fills inadmissible data.

The system will provide:

- *Online user help*
- *System usage documentation*

2.7 Assumptions and Dependencies

AS - 1. Users have stable internet connectivity

AS - 2. Users possess institute-provided Gmail accounts

AS - 3. Standard web browsers are available

AS - 4. Email notification services are operational

3. External Interface Requirements

3.1 User Interfaces

*The system shall provide a **web-based user interface** for all user roles, ensuring usability and consistency across devices.*

UI - 1. The website shall start with a landing page . The landing page will provide the option to in using the Institute mail.

UI - 2. Instruction will be provided to the user on top of forms to be filled.

UI - 3. There will be alerts and pop ups in case the user makes a mistake in filling the form.

UI - 4. The interface will be responsive for all screen sizes as much as possible to provide the users a seamless experience.

3.2 Hardware Interfaces

N/A

3.3 Software Interfaces

- *Browsers: Mozilla Firefox 30+, Google Chrome 27.0+ are the preferred browsers.*
- *Operating System: Android, Windows 7, 8, 10, Mac OS, Linux distributions.*

3.4 Communications Interfaces

- *Communication protocols: **HTTPS***
- *Data formats: **JSON***

4. System Features

4.1 User Authentication and Authorization

4.1.1 Description and Priority

Provides secure login and role-based access control.
*Priority: **High***

4.1.2 Stimulus/Response Sequences

- *User attempts login using institute Gmail*
- *System authenticates credentials*
- *System grants access based on role*

4.1.3 Functional Requirements

- *FR-AUTH-01: The system shall allow users to log in using institute Gmail accounts.*
- *FR-AUTH-02: The system shall authenticate users securely.*
- *FR-AUTH-03: The system shall enforce role-based access control.*

4.2 Student Academic Management

4.2.1 Description and Priority

Allows students to manage / submit academic information and other co-curricular activities.
Priority: ***High***

4.2.2 Stimulus/Response Sequences

- *Students enter academic data and a description of the co - curricular activities.*
- *Student submits data*
- *System records submission status*

4.2.3 Functional Requirements

- *FR-STU-01: The system shall allow students to view and update personal profiles.*
- *FR-STU-02: The system shall allow students to enter academic details such as . courses, grades, and semester performance.*
- *FR-STU-03: The system shall allow students to update academic data*
- *FR-STU-04: The system shall allow students to enter a description of their . co - curricular activities.*
- *FR-STU-05: The system shall allow students to submit academic data for verification.*
- *FR-STU-06: The system shall display verification status (pending/approved/rejected)*
- *FR-STU-07: The system shall display faculty remarks when data is rejected***.**

4.3 Faculty Verification and Review

4.3.1 Description and Priority

Supports academic verification by faculty members.
Priority: ***High***

4.3.2 Stimulus/Response Sequences

- *Faculty views submitted data*
- *Faculty approves or rejects records*
- *System notifies student*

4.3.3 Functional Requirements

- *FR-FAC-01: The system shall allow faculty to view assigned students.*
- *FR-FAC-02: The system shall allow faculty to review submitted academic data.*
- *FR-FAC-03: The system shall allow faculty to approve verified records.*
- *FR-FAC-04: The system shall allow faculty to reject records with remarks.*

4.4 Notifications and Alerts

4.4.1 Description and Priority

Provides automated communication to users.
Priority: ***Medium***

4.4.2 Functional Requirements

- *FR-NOT-01: The system shall send email notifications on approval or rejection.*
- *FR-NOT-02: The system shall notify students of discrepancies.*

4.5 Chatbot-Based Academic Guidance

4.5.1 Description and Priority

Provides academic insights and guidance.
Priority: ***Medium***

4.5.2 Functional Requirements

- *FR-BOT-01: The system shall analyze academic history to identify weaknesses.*
- *FR-BOT-02: The system shall suggest electives based on past performance.*

4.6 Admin Management

4.6.1 Description and Priority

Supports administrative control of the system.
Priority: ***High***

4.6.2 Functional Requirements

- *FR-ADM-01: The system shall allow admins to add faculty members.*
- *FR-ADM-02: The system shall allow admins to assign roles.*
- *FR-ADM-03: The system shall allow admins to manage access rights.*
- *FR-ADM-04: The system shall maintain system records***.**

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- *The system shall respond to user actions within* ***2 seconds*** *for at least* ***95% of requests*** *under normal academic load.*
- *The system shall support* ***concurrent access by a minimum of 500 users*** *without performance degradation.*
- *The system shall ensure that academic data submission and verification operations are completed without data loss or inconsistency.*
- *The system shall generate academic reports within* ***5 seconds*** *for standard report sizes.*

5.2 Safety Requirements

- *No specific safety requirements are applicable, as the system does not interact with physical hardware or safety-critical components.*
- *The system shall ensure that accidental data loss due to system failure is minimized through regular backups.*

5.3 Security Requirements

- *The system shall provide **secure authentication** mechanisms for all users.*
- *The system shall restrict access to system features based on **role-based authorization** (Student, Faculty Advisor, Admin).*
- *The system shall ensure **data confidentiality and integrity** during storage and transmission.*
- *The system shall use **HTTPS** for all client–server communication.*
- *The system shall prevent unauthorized access to academic records.*
- *The system shall maintain **audit logs** for critical actions such as submissions, approvals, and rejections.*

5.4 Software Quality Attributes

The system shall emphasize the following quality attributes:

- ***Usability**:*
  The system shall provide an intuitive and user-friendly interface accessible through standard web browsers.
- ***Reliability**:*
  The system shall ensure consistent availability during academic periods and recover gracefully from failures.
- ***Maintainability**:*
  The system shall be modular in design to allow easy updates, bug fixes, and feature enhancements.
- ***Scalability**:*
  The system shall support future increases in users, academic data, and feature additions without major architectural changes.

5.5 Business Rules

- *Only faculty members are authorized to approve or reject submitted academic records.*
- *Students shall be allowed to modify academic data **only before submission** for verification.*
- *Role creation and assignment shall be restricted to system administrators.*
- *Approved academic records shall be locked from further modification unless reopened by authorized personnel.*

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6. Other Requirements

- *The system shall maintain database consistency and integrity.*
- *The system shall support logging and audit trails for academic actions.*

Appendix A: Glossary

- ***FA:*** *Faculty Advisor*
- ***Admin:*** *System administrator*
- ***PMS :*** *Performance Monitoring System*
- ***HTTPS*** *: Hyper Text Transfer Protocol Secure*